

Module 1

Session 1

1) Name 5 IT job roles and write one line about what each role does.

Ans=

- Software Developer: Writes, tests, and maintains code to build applications or systems.
- Web Developer: Creates and maintains websites and web applications using HTML, CSS, JavaScript, and backend technologies.
- Database Administrator (DBA): Manages, secures, and optimizes databases to store and retrieve data efficiently.
- Network Administrator: Installs, configures, and maintains an organization's computer networks and ensures their security and uptime.
- QA Tester: Tests software to find bugs and ensures it works as expected before release.

2) List 3 product companies and 3 service companies with one example of work each does.

Ans=

Product companies:

- Microsoft: Develops and sells software products like Windows OS and Microsoft 365.
- Adobe: Creates creative and design software like Photoshop and Premiere Pro.
- Oracle: Builds database software and enterprise applications like Oracle Database and ERP systems.

Service companies:

- Tata Consultancy Services (TCS): Provides IT consulting, software development, and outsourcing services to clients.
- Infosys: Offers digital transformation, application development, and maintenance services globally.
- Wipro: Delivers IT services including cloud, cybersecurity, and infrastructure management for businesses.

3) Full forms and one-line use of: GitHub, Slack, Jira, IDE, API.

Ans=

- GitHub: "Git" + "Hub"; used for hosting and collaborating on code repositories using Git.
- Slack: Informally "Searchable Log of All Conversation and Knowledge"; used for team chat and collaboration.
- Jira: No official full form; used for tracking bugs, tasks, and agile project management.
- IDE: Integrated Development Environment; used to write, debug, and run code in one application.
- API: Application Programming Interface; used to allow different software applications to communicate and share data.

4) Name any 5 programming languages and mention one popular use case for each.

Ans=

- Python: Used for data science, AI/ML, and automation scripts.
- JavaScript: Used for building interactive websites and web applications.
- Java: Used for enterprise applications, Android apps, and large-scale backend systems.
- C#: Used for building Windows desktop apps and ASP.NET web applications.
- SQL: Used for querying and managing relational databases.

5) Difference between a developer and a tester (2 lines).

Ans=

- A developer writes and builds the software code and features.
- A tester checks the software for bugs and ensures it meets requirements before release.

Session 2

1) Three main layers of software architecture.

Ans=

- Presentation layer (UI layer): Shows the interface to the user and handles input/output (buttons, forms, screens).
- Application layer (business logic layer): Contains the core rules and processing (calculations, validations, workflows).
- Data layer (data access layer): Manages storage and retrieval of data from databases or files.

2) Define client and server with one real-life example.

Ans=

- Client: A program or device that requests services or data from another system (e.g., your browser asking for a webpage).
- Server: A program or machine that provides services or data in response to client requests (e.g., a web server sending HTML pages).

Example: When you open Gmail in Chrome, your browser is the client and Google's mail servers are the server.

3) Two differences each between desktop apps, web apps, and mobile apps.

Ans=

Desktop vs Web:

- Desktop apps are installed on a specific OS; web apps run in a browser and are OS independent.
- Desktop apps are updated locally; web apps are updated centrally on the server.

Web vs Mobile:

- Web apps run in browsers using HTML/CSS/JS; mobile apps are built for Android/iOS using native SDKs.

- Mobile apps have deep hardware access and are on app stores; web apps have limited hardware access and use URLs.

Desktop vs Mobile:

- Desktop apps are for large screens with keyboard/mouse; mobile apps are for touch and small screens.
- Desktop apps can use more CPU/RAM; mobile apps must optimize for battery and memory.

4) Name any 3 databases and mention whether each is SQL or NoSQL.

Ans=

- MySQL: SQL (relational database).
- PostgreSQL: SQL (advanced relational database).
- MongoDB: NoSQL (document-oriented database).

Session 3

1) Full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

Ans=

- HTTP: HyperText Transfer Protocol; used to transfer web pages between browser and server.
- HTTPS: HyperText Transfer Protocol Secure; same as HTTP but with encryption for secure communication.
- FTP: File Transfer Protocol; used to upload and download files between client and server.
- SMTP: Simple Mail Transfer Protocol; used to send emails from client to mail server or between mail servers.

2) Difference between IPv4 and IPv6 (2 lines).

Ans=

- IPv4: Uses 32-bit addresses (e.g., 192.168.1.1) and has about 4.3 billion unique addresses.
- IPv6: Uses 128-bit addresses (e.g., 2001:0db8::1) and provides a vastly larger address space for future growth.

3) Run nslookup google.com and write the IP address.

Ans=

Command: nslookup google.com

Example output (your IP may differ): Address: 142.250.196.78

4) HTTP status codes: 200, 301, 404, 500, 403.

Ans=

- 200 OK: Request successful; server returned the requested data.
- 301 Moved Permanently: Resource has moved to a new URL permanently.
- 404 Not Found: Requested page or resource does not exist on the server.
- 500 Internal Server Error: Server encountered an unexpected condition and failed to fulfill the request.
- 403 Forbidden: Server understands the request but refuses to authorize it (access denied).

5) DNS hierarchy with www.google.com.

Ans=

- Root domain (.): Top of DNS tree; root servers know about TLDs.
- TLD (Top-Level Domain): Like .com, .org; root points to TLD servers.
- Domain: Like google.com; TLD servers point to domain's authoritative name servers.
- Subdomain: Like www in www.google.com; part of the domain used to organize services.

For www.google.com: Root → .com TLD → google.com domain → www subdomain.

Session 4

1) 6 phases of SDLC in order with one line about each.

Ans=

1. Requirement Analysis: Gather and understand what the software must do.
2. Design: Plan architecture, database, and UI/UX based on requirements.
3. Implementation (Coding): Write actual code to build the software.
4. Testing: Find and fix bugs; ensure software meets requirements.
5. Deployment: Release the software to production for end users.
6. Maintenance: Provide updates, fixes, and improvements after deployment.

2) 3 differences between Agile and Waterfall models.

Ans=

- Approach: Waterfall is linear and sequential; Agile is iterative and incremental.
- Flexibility: Waterfall is rigid once a phase is completed; Agile allows changes at any time.
- Delivery: Waterfall delivers at the end of the project; Agile delivers working software in short cycles (sprints).

3) 3 main Scrum roles and one-line responsibility.

Ans=

- Product Owner: Defines and prioritizes the product backlog to maximize value.
- Scrum Master: Facilitates Scrum process, removes obstacles, and coaches the team.
- Development Team: Self-organizing team that designs, builds, and tests the product.

4) What is a Sprint in Scrum? Mention its typical duration.

Ans=

- A Sprint is a time-boxed iteration in Scrum where a set of backlog items is developed into a potentially shippable product increment.
- Typical duration: 2 to 4 weeks (most commonly 2 weeks).

5) Jira board columns: To Do, In Progress, Done, Backlog.

Ans=

- Backlog: List of all planned tasks and features not yet scheduled for a sprint.
- To Do: Tasks selected for the current sprint but not yet started.
- In Progress: Tasks currently being worked on by the team.
- Done: Tasks completed, tested, and ready for release.

Session 5

1) Git commands.

Ans=

- Initialize a repo: `git init`
- Stage all files: `git add .`
- Commit with a message: `git commit -m "Your commit message here"`
- Push to remote: `git push origin <branch-name>` (e.g. `git push origin main`)
- Pull latest changes: `git pull origin <branch-name>` (e.g. `git pull origin main`)

2) Difference between git fetch and git pull (2 lines).

Ans=

- `git fetch` downloads new commits and branches from remote but does not merge them into your current branch.
- `git pull` does a fetch and then automatically merges the changes into your current branch.

3) Create GitHub repo, push a small text file, share link.

Ans=

Steps:

1. Create a new repo on GitHub (via website)
2. On your local machine:

`git init`

`echo "Hello Git" > test.txt git add test.txt git commit -m "Initial commit
with test file" git branch -M main git remote add origin`

`https://github.com/your-username/your-repo-name.git git push -u origin
main`

Repo link example `https://github.com/your-username/your-repo-name`

4) Difference between a Fork and a Pull Request (2 lines).

Ans=

- A Fork is your own copy of someone else's repository, created on GitHub so you can experiment freely.
- A Pull Request (PR) is a request to merge your changes (from your fork or branch) back into the original repository.

5) Create branch feature-test, make one commit, merge to main; write commands.

Ans=

Commands:

Create and switch to new branch git

checkout -b feature-test

Make some change

echo "Feature test" > feature.txt git

add feature.txt git commit -m "Add
feature test file"

Switch back to main git

checkout main

Merge feature-test into main git

merge feature-test

(Optional) Delete the feature branch

git branch -d feature-test